

9

NUMBER AND ALGEBRA

THE NUMBER PLANE

Any place on the Earth's surface can be described using 2 numbers called **coordinates**. We can describe location using lines of **latitude** and **longitude**. Latitude measures the angle north or south from the equator, the imaginary horizontal line that runs across the middle of the Earth. Longitude measures the angle east or west from the Prime Meridian, the imaginary vertical line that runs through Europe and the Greenwich Observatory near London, England. The coordinates of latitude and longitude can be used to describe any point on Earth.



Chapter outline

	Working mathematically				
9.01 Location using coordinates	U	F		R	C
9.02 The number plane	U	F		R	C
9.03 Quadrants of the number plane	U	F		R	C
9.04 Graphing tables of values	U	F		R	C
9.05 Completing tables of values	U	F			
9.06 Graphing linear equations	U	F		R	C

Wordbank

coordinates The 2 values that give the position of a point on a number plane

linear equation A formula whose graph is a straight line

number plane A grid that is made up of 2 number lines, meeting at right angles

ordered pair Another name for coordinates

origin The centre point of a number plane, with coordinates (0, 0)

quadrant One quarter of the number plane

table of values A table of ordered pairs of numbers, usually following a formula and which can be graphed on a number plane

***x*-axis** The horizontal number line on the number plane

***y*-axis** The vertical number line on the number plane

Quiz
Wordbank 9

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (3):

9.03 The number plane

9.06 Graphing linear equations • Linear functions

PhET interactives (3):

9.06 Function builder: Basics • Function builder • Graphing lines

Quizzes (5):

- Wordbank 9
- SkillCheck 9
- Mental skills 9
- Language of maths 9
- Test yourself 9

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9.02, 9.03 The number plane

Worksheets (13):

9.02 1 cm grid paper • Plane grid 1

9.03 Plane grid 2 • The number plane • Number plane writing activity • Number plane review

9.04 Graphing tables of values

9.05 Tables of values

9.06 Graphing linear equations • Number plane grid paper • A page of lines • A page of number planes • Plane grid 2

Mind map The number plane

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9.03 Girt by sea • Quadrilateral person • Coordinates code puzzle

9.06 Matching linear equations • Graphing functions

Language of maths Number plane crossword

Presentation (1):

9.03 Graph coordinates

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9.06 Graphing straight lines: Finding intercepts

Spreadsheet (1):

9.06 Linear function plotter

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

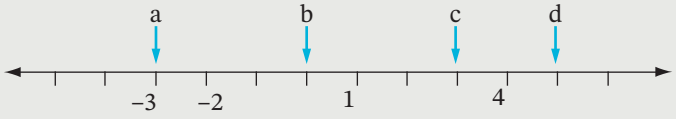
- use grid coordinates to describe the position of a place
- learn about the Cartesian plane and coordinate system
- locate and plot points on the Cartesian plane, including all 4 quadrants
- graph points from a table of values, including points on a line
- complete a table of values for a linear equation and graph the linear equation on a number plane

SkillCheck ANSWERS ON P. 591

Quiz
SkillCheck 9

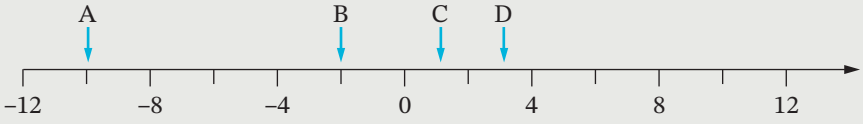
9

1 Write the integer shown by each arrow on this number line.

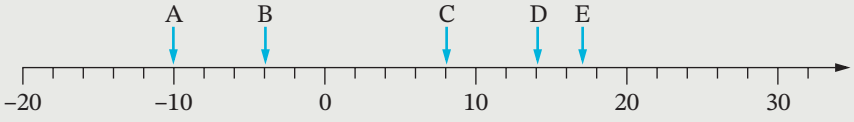


2 Write the integer shown by each arrow on each number line.

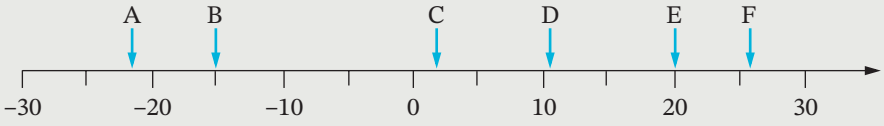
a



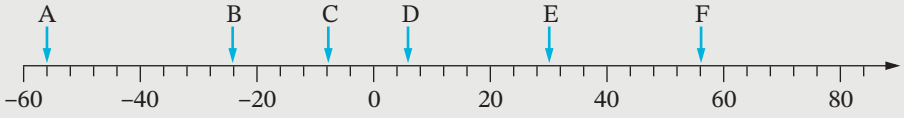
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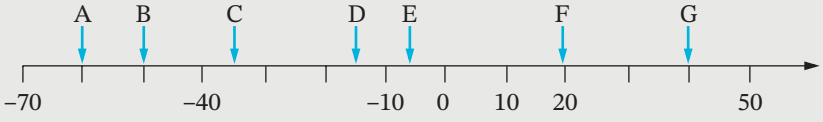
c



d



e



3 Mark each set of numbers as dots on a number line.

a −2, 4, 8, −3, 0

b the integers between −5 and 1

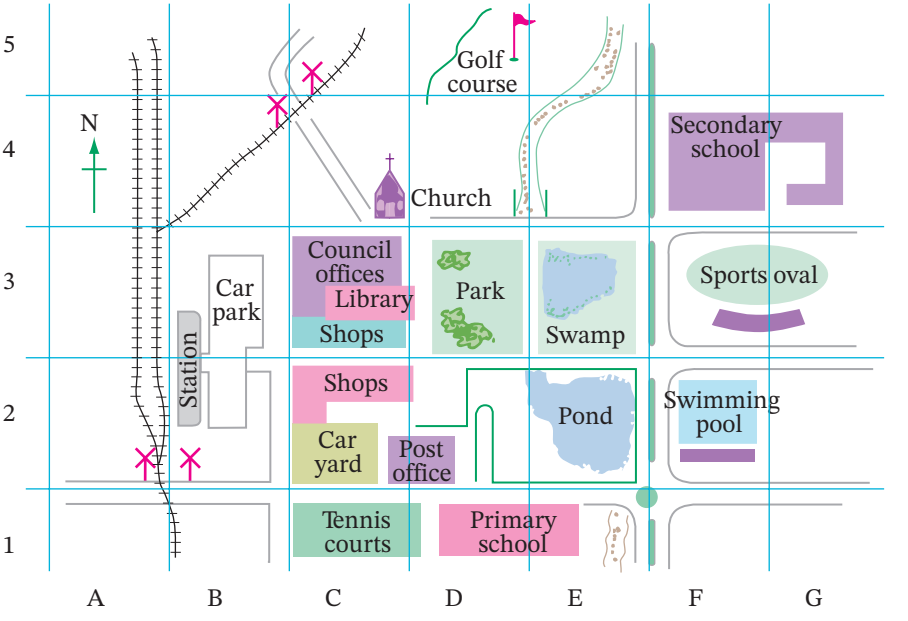
c the negative and positive integers between −10 and 4 that are multiples of 3

d −1.5, 5.5, 2.5, −4.5, −2.5

9.01 Location using coordinates

To locate a position on a map or theatre seating plan, we usually use a grid system of 2 coordinates – one for the horizontal and one for the vertical. On a map, this may be a letter and a number, on a theatre plan, it may be a row letter and a number.

This map of a country town uses a coordinate grid system where every position can be described by a column letter and row number. For example, the position of the golf course is D5, because it is in the square in column D, row 5.



EXERCISE 9.01 ANSWERS ON P. 591

Location using coordinates U F R C

1 Use the town map to write the position of: C

a the swamp **b** the library **c** the church

d the secondary school **e** the sports oval **f** the council offices

2 On the town map, describe what can be found at: C

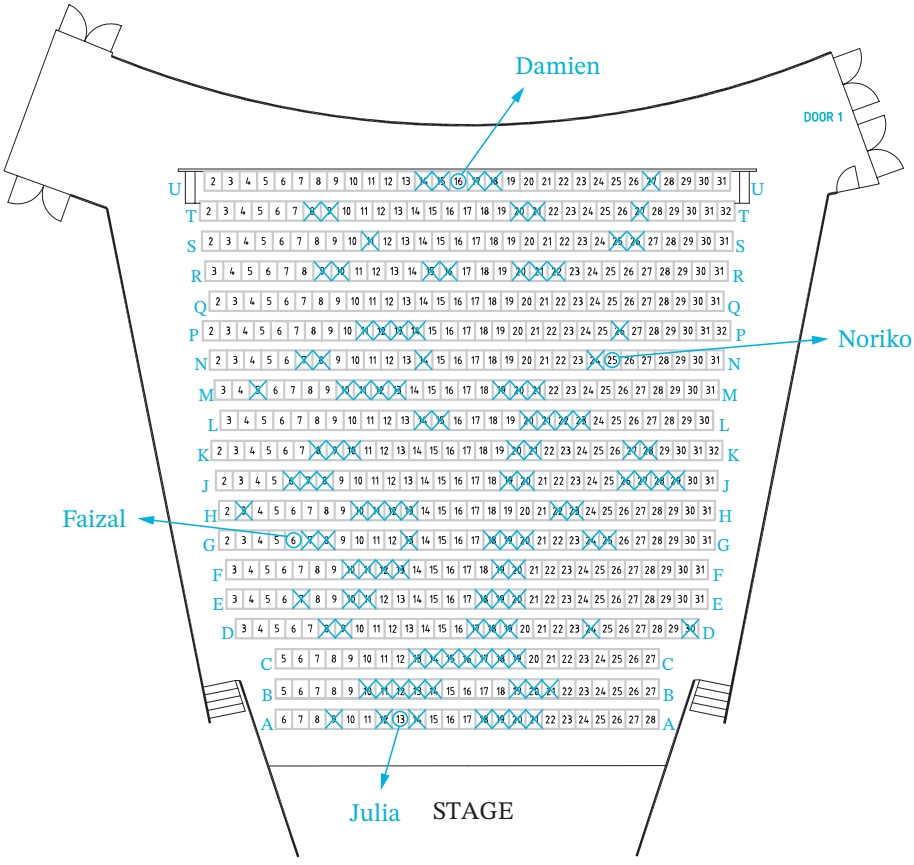
a F2 **b** D1 **c** D3

d E2 **e** B3 **f** C1

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3 This map shows the seating plan for the Drama Theatre at the Sydney Opera House. Each seat in the theatre has a row number and a position number, for example, seat A17 is in the middle of the front row.



Write the seat location of each person. C

a Julia **b** Faizal **c** Noriko **d** Damien

4 How many rows are there in the Drama Theatre?

5 Are there more seats in the front rows or the back rows? R C

6 The seats marked with a cross have already been booked. State whether each seat has been booked:

a D4 **b** H15 **c** R11 **d** T9

7 Lara, Zoe and Asha want to sit together at the Drama Theatre. Give 2 sets of possible locations where they could sit. R

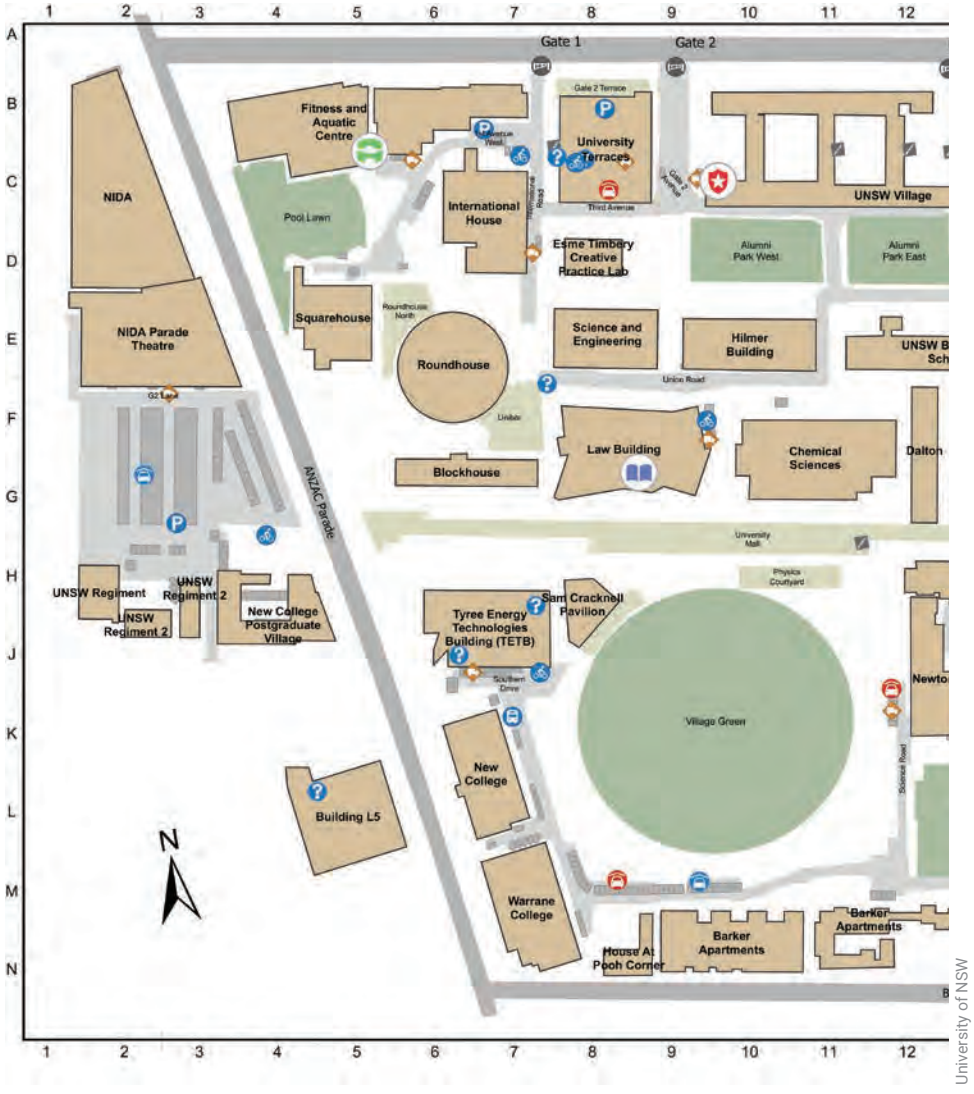
8 What type of pattern would be seen if all the people sitting in: C

a a seat with number 15 stood up?

b a seat with letter K stood up?

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9 This is part of a map of a university campus.



University of NSW

Name the building or feature that can be found at: C

a E6 **b** L5 **c** E2

d C6 **e** D12 **f** F9

10 From the university map, write the location of: C

a Warrane College **b** Chemical Sciences

c Western Campus carpark **d** Squarehouse

The number plane

9.02

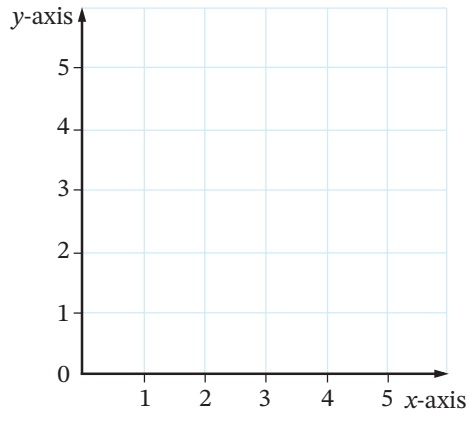
A **number plane** is a grid made from a **horizontal** number line called the ***x*-axis**, and a **vertical** number line called the ***y*-axis**. **Axis** means 'line', and the plural of axis is 'axes'.

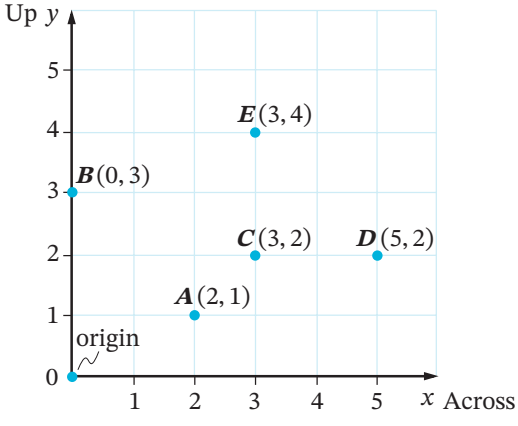
The number plane is also called a **Cartesian plane**, named after **René Descartes** (pronounced 'Ren-ay Day-cart'), a French mathematician and philosopher who developed the idea.

Points on the number plane are located using a pair of **coordinates**, for example (3, 2) means 3 on the *x*-axis ('3 across') and 2 on the *y*-axis ('2 up'). A pair of coordinates is also called an **ordered pair**.

On the number plane on the right:

- *A* is the point (2, 1)
- *B* is the point (0, 3).





One way to remember that '*x* is across' is because 'X is a cross'!

Skillsheet
The number plane

Puzzle
Big top employee

9.02

Coordinates on the number plane

- In any ordered pair, the first number is called the ***x*-coordinate** and the second number is called the ***y*-coordinate**.
- The point (0, 0) is called the **origin**.
- To locate a point, always start from (0, 0): the first coordinate tells you how far to move across, the second tells you how far to move up.

(2, 1) means (2 across, 1 up)

- start from (0, 0)
- move 2 units across
- then 1 unit up

(0, 3) means (0 across, 3 up)

- start from (0, 0)
- do not move across
- move 3 units up

Note that order is important with coordinates: (0, 3) is not the same as (3, 0).

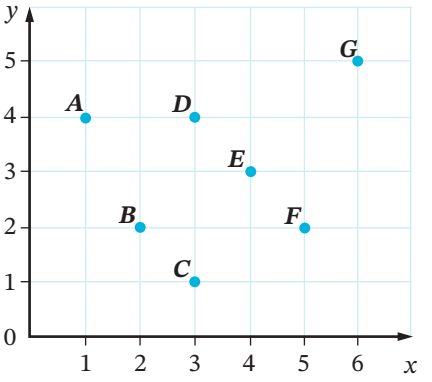
EXERCISE 9.02 ANSWERS ON P. 591

The number plane UFRC

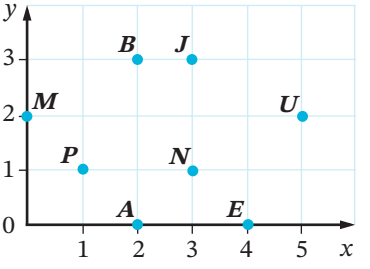
Worksheet
1 cm grid paper

Some questions in this exercise require you to draw a number plane. You may want to download the worksheet '1 cm grid paper' to do this.

1 Write the coordinates of each point shown on this number plane. C



2 Write the coordinates of the 8 points shown on this number plane. C



3 Draw a number plane with the numbers from 0 to 6 on each axis.

a Start at the point (1, 1) and draw a line joining it to (4, 1).

b Join (4, 1) to (6, 2), then continue through (3, 2) → (1, 1) → (1, 4) → (3, 5) → (6, 5) → (4, 4) and finish at (1, 4).

c Now join (3, 2) to (3, 5).

d Join (4, 1) to (4, 4).

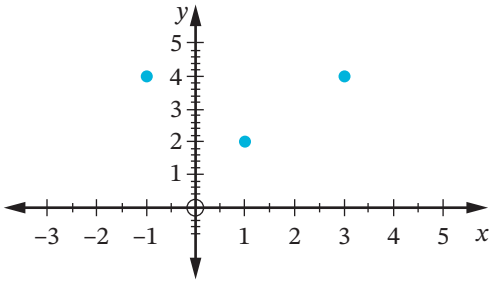
e Join (6, 2) to (6, 5).

f What have you drawn?

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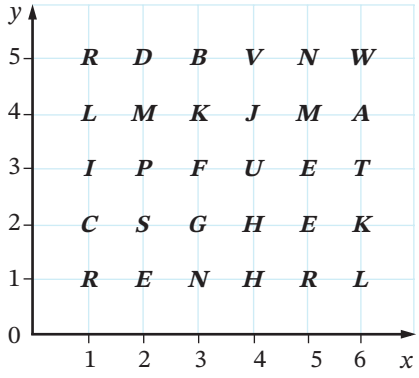
4 Aiden plots these 3 points on the number plane.



He wants to add a 4th point to form a parallelogram when you join the points. Which point could Aiden plot? Select the correct answer **A**, **B**, **C** or **D**. R

A (−3, 2) **B** (3, 2) **C** (5, 1) **D** (−1, 2)

5 Write the letter shown at each point on the number plane on the right, to spell some words. C



a (3, 1), (4, 3), (2, 4), (3, 5), (5, 2), (1, 5)

b (2, 3), (6, 1), (6, 4), (5, 5), (2, 1)

c (6, 5), (6, 4), (6, 3), (5, 3), (5, 1)

d (2, 4), (5, 2), (6, 3), (5, 1), (5, 3)

e (3, 1), (4, 3), (5, 1), (2, 2), (2, 1)

f (4, 3), (2, 4), (2, 3), (1, 3), (1, 1), (2, 1)

6 **a** Draw a number plane with the *x*-axis going from 0 to 20 and the *y*-axis going from 0 to 15.

b Plot the following points and connect them as you go. Colour appropriately.

- (0, 9), (8, 11), (10, 14), (15, 10), (16, 12), (17, 9), (19, 7), (19, 5), STOP
- (19, 7), (15, 5), (11, 9), (13, 5), (11, 0), (15, 4), (19, 5), STOP
- (17, 6), (13, 7), STOP
- (12, 7), (10, 7), (11, 3), (7, 7), (3, 7), (5, 8), (2, 8), (0, 9), STOP
- (3, 9), STOP

c What did you draw?

7 **a** Draw a number plane or use the worksheet 'Plane grid 1'. On the *x*-axis, mark 0 to 11. On the *y*-axis mark 0 to 16.

b Plot the following points, joining them as you go.

(1, 13), (3, 14), (4, 15), (5, 15), (6, 14), (6, 13), (4, 10), (7, 11), (9, 10), (10, 9), (11, 6), (11, 4), (8, 8), (9, 2), (9, 1), (7, 1), (8, 2), (7, 6), (6, 6), (7, 4), (7, 1), (5, 1), (6, 2), (6, 4), (5, 6), (3, 7), (2, 9), (2, 10), (4, 13), (1, 13)

c Draw an eye at (4, 14)

d What did you draw?

9.02

Worksheet
Plane grid 1

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Did you know?

Global positioning system GPS



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The global positioning system (GPS) originated in the 1960s with the need for the Russians and the United States Navy to track nuclear submarines. It is a space-based radio navigation system owned by the US Government and is freely available for all to use. Every smartphone and new car has a GPS application that allows you to find the position of any place in the world. The system requires a clear line of sight to 4 or more of the 30 GPS satellites circling the world. Other countries are now developing their own GPS that can precisely pinpoint a location to within 30 centimetres.

Research how the GPS receiver knows where you are.

9.03 Quadrants of the number plane

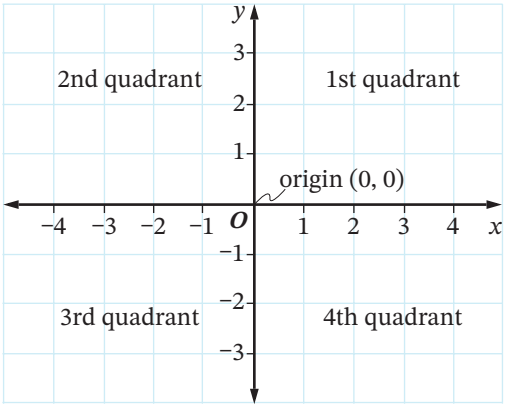
Skillsheet
The number plane

Worksheets
Plane grid 2
The number plane
Number plane writing activity
Number plane review

We can extend the number plane to use both positive and negative coordinates, by including negative numbers on the x- and y-axes.

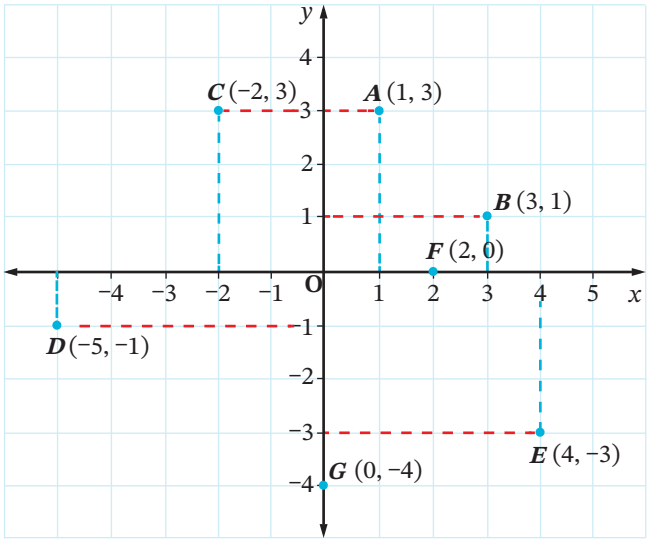
The number plane is divided evenly into 4 regions called **quadrants**.

The centre point (0, 0) is called the **origin**.



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A few points are labelled on this number plane as examples.

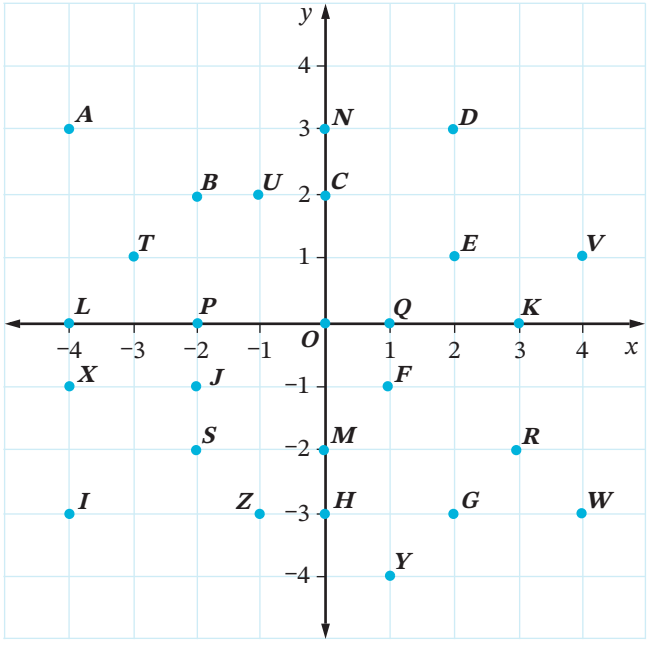


EXERCISE 9.03 ANSWERS ON P. 591

Quadrants of the number plane UFRC

Some questions in this exercise require you to draw a number plane. You may want to use the 'Plane grid 2' worksheet for this.

1 Write the coordinates of each of the 26 points shown on the number plane. C



Puzzles
Girt by sea
Quadrilateral person
Coordinates code puzzle

9.03

Video
The number plane

Presentation
Graph coordinates

Worksheet
Plane grid 2

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2 **a** Draw a number plane with both axes extending from –6 to 6.

b Plot and label each point on the number plane. C

$A(3, 1)$ $B(-4, 3)$ $C(-3, 4)$ $D(-2, -2)$

$E(0, -2)$ $F(1, -5)$ $G(4, -4)$ $H(-3, 0)$

$I(-6, 0)$ $J(-1, -6)$ $K(-3, -4)$ $L(0, 5)$

$M(6, 2)$ $N(-3, 5)$ $P(5, -5)$ $Q(-1, 6)$

3 List all of the points from question **2** that are: C

a in the 1st quadrant **b** in the 2nd quadrant **c** in the 3rd quadrant

d in the 4th quadrant **e** on the border of 2 quadrants.

4 In which quadrant would you find each point? C

a $(3, -5)$ **b** $(-2, -4)$ **c** $(-8, 1)$

5 Copy and complete this table by writing + or –. R

Points in:	*x*-coordinate	*y*-coordinate
1st quadrant	+	
2nd quadrant	–	
3rd quadrant		
4th quadrant		

6 Write the coordinates of the point that is:

a 3 units above $(1, -2)$ **b** $1\frac{1}{2}$ units to the right of $(-4, -1)$

7 **a** Draw a number plane, with both axes extending from –3 to 3.

b Plot and label each point.

$A(0, 1)$ $B(2, -1)$ $C(-1, 2)$ $D(-2, 3)$ $E(3, -2)$ $F(1, 0)$

c What do you notice about the points? Check this with your ruler.

d Points that lie on the same straight line are called **collinear**. Write down the coordinates of any 4 collinear points on your number plane. C

8 Plot and label the points $P(1, -1)$, $Q(2, 1)$, $R(-1, 2)$, $S(-2,0)$ on a number plane. What shape is *PQRS*? Select the correct answer **A**, **B**, **C** or **D**. R

A rectangle **B** square **C** rhombus **D** parallelogram

9 **a** Draw a number plane with both axes extending from –10 to 10.

b Start at the first point and join the points in the order given:

- (4, 0), (5, 1), (8, 1), (9, 0), (9, –3), (8, –4), (5, –4), (4, –3), (4, 0), STOP
- (–2, 0), (–3, 1), (–6, 1), (–7, 0), (–7, –3), (–6, –4), (–3, –4), (–2, –3), (–2, 0), STOP
- (7, –2), (4, 5), (1, –1), (–2, 5), (–5, –2), STOP
- (3, 6), (5, 6), (4, 5), (–2, 5), (–2, 6), (–4, 6), (–4, 4), (–3, 4), STOP
- (1, 1), (1, –3), STOP
- (0.5, 1), (1.5, 1), STOP
- (0.5, –3), (1.5, –3), STOP

c What did you draw? C

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Investigation

Number plane picture puzzles

- Draw a number plane from –10 to 10 on both x- and y-axes.
- Create your own number plane drawing using all 4 quadrants.
- Write out the set of points so that someone else could draw it.

Graphing tables of values 9.04

Ordered pairs of coordinates can also be written in a **table of values**.



Worksheet
Graphing tables of values

x	y
–3	–2
–2	–1
0	2
1	0

means
(–3, –2)
(–2, –1)
(0, 2)
(1, 0)

x	0	1	2
y	5	7	3

means
(0, 5)
(1, 7)
(2, 3)

EXERCISE 9.04 ANSWERS ON P. 592

Graphing tables of values UFRC

1 Write down the ordered pairs shown in each table of values. C

a

x	y
0	2
2	3
4	4
7	9

b

x	y
–2	–1
–1	3
2	4
5	–2

c

x	0	1	2
y	2	5	8

d

x	–4	–2	1	5
y	–1	–3	0	2

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2 Plot the coordinates shown in each table of values on a separate number plane, with both axes extending from –4 to 4. Use a ruler to join each set of points, and describe the pattern of points. C

a

x	0	1	2	3
y	2	2	2	2

b

x	–1	0	1	2
y	–1	–1	–1	–1

c

x	3	–1	1	–3
y	1	1	1	1

d

x	–3	2	0	4
y	–3	–3	–3	–3

3 Graph this table of values on a number plane, with both axes extending from –4 to 4, and describe the pattern of points. C

x	–2	–1	0	1
y	0	0	0	0

4 Graph each table of values on a number plane. Use a ruler to join each set of points and describe their pattern. C

a

x	1	1	1	1
y	–2	–1	0	1

b

x	–2	–2	–2	–2
y	–2	–1	0	1

c

x	3	3	3	3
y	–2	1	–1	3

d

x	–1	–1	–1	–1
y	–3	–1	0	2

5 Graph each table of values on a number plane, join each set of points and describe their pattern. C

a

x	–2	–1	1	2
y	–1	0	2	3

b

x	–2	0	1	3
y	–4	–2	–1	1

c

x	–4	–2	2	4
y	–1	0	2	3

d

x	–3	–1	0	1
y	–3	1	3	5

6 For the tables of values in question **5**, what happens to the y-values as the x-values increase? R C

7 A line is drawn through the points (0, 4), (2, 0) and (–1, 6). Which of the following points also lies on the line? Select the correct answer **A**, **B**, **C** or **D**.

A (1, –1) **B** (1, 2) **C** (–2, 7) **D** (3, –3)

8 Graph each table of values on a separate number plane and describe the pattern of the points. C

a

x	–3	–2	–1	0	1	2	3
y	9	4	1	0	1	4	9

b

x	–2	–1	0	1	2
y	–8	–1	0	1	8

c

x	–2	–1	0	1	2	3	4	5	6
y	0	–2	0	2	0	–2	0	2	0

9 This table shows the height of a tree as it ages. Plot each point on a number plane, where the x-coordinate is the age of the tree and the y-coordinate is the height of the tree. After plotting each (age, height) point, explain whether there is a pattern between the age and height of the tree. R C

Age (years)	5	10	15	20	25
Height of tree (metres)	10	12.5	17.5	21	26



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Foundation Standard Complex

Quiz
Mental skills 9

Mental skills 9: Maths without calculators ANSWERS ON P. 593

Simplifying multiplication by factorising

Sometimes, one big multiplication may be split into many simpler little multiplications if we break each term into 2 'easier' factors. We use the property that numbers can be multiplied in any order.

1 Study each example.

a
$$\begin{aligned} 15 \times 6 &= 3 \times 5 \times 3 \times 2 \\ &= 3 \times 3 \times 5 \times 2 \\ &= 9 \times 10 \\ &= 90 \end{aligned}$$

b
$$\begin{aligned} 30 \times 22 &= 3 \times 10 \times 2 \times 11 \\ &= 3 \times 2 \times 10 \times 11 \\ &= 6 \times 11 \times 10 \\ &= 660 \end{aligned}$$

c
$$\begin{aligned} 24 \times 8 &= 6 \times 4 \times 2 \times 4 \\ &= 4 \times 4 \times 6 \times 2 \\ &= 16 \times 6 \times 2 \\ &= 96 \times 2 \\ &= 192 \end{aligned}$$

d
$$\begin{aligned} 14 \times 36 &= 7 \times 2 \times 6 \times 6 \\ &= 6 \times 7 \times 6 \times 2 \\ &= 42 \times 6 \times 2 \\ &= 252 \times 2 \\ &= 504 \end{aligned}$$

Look for different pairs of numbers that you can multiply easily and rearrange.
Note: For each question, there are many different possible ways of arriving at the a correct result.

2 Now evaluate each product by factorising first.

a	30×24	**b**	18×27	**c**	25×33	**d**	16×8
e	28×20	**f**	12×18	**g**	21×9	**h**	36×12
i	16×35	**j**	22×28	**k**	9×15	**l**	27×25

Completing tables of values

9.05

We will now use a rule or formula to create a **table of values** that will help us graph points on the number plane.

Worksheet
Tables of values
9.05

Example 1

Complete this table of values using the formula $v = 2u + 1$.

u	−2	−1	0	1	2	6	9
v							

Solution

Substitute each value of u in the table into the formula $v = 2u + 1$ to find the value of v.

- When $u = -2$, $v = 2 \times (-2) + 1 = -3$
- When $u = -1$, $v = 2 \times (-1) + 1 = -1$
- When $u = 0$, $v = 2 \times 0 + 1 = 1$
- When $u = 1$, $v = 2 \times 1 + 1 = 3$
- When $u = 2$, $v = 2 \times 2 + 1 = 5$
- When $u = 6$, $v = 2 \times 6 + 1 = 13$
- When $u = 9$, $v = 2 \times 9 + 1 = 19$

Completing the table, we have:

u	−2	−1	0	1	2	6	9
v	−3	−1	1	3	5	13	19

EXERCISE 9.05 ANSWERS ON P. 593

Completing tables of values U F

EXAMPLE 1

1 Complete each table of values given the formula.

a $y = x - 1$

x	3	7	−1	−8	4	5
y					3	

b $q = 3p$

p	8	−3	2	5	10	−4
q				15		

c $k = h \div 2$

h	12	8	−2	−6	0	7
k	6					

d $y = x + 7$

x	−2	11	7	0	13	−9
y		18				

e $b = 2a$

a	8	−1	−5	0	4	2
b			−10			

f $y = x - 3$

x	10	−4	7	5	−8	11
y	7					

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2 Given the rule $y = -x - 4$, what is the value of y when $x = -3$? Select the correct answer **A**, **B**, **C** or **D**.

A -1 **B** -7 **C** 7 **D** 12

3 Complete each table of values given the formula.

a $p = 5m - 1$

m	3	10	1	4	8	−6
p		49				

b $q = 3p - 2$

p	4	−1	7	10	2	−6
q			19			

c $t = 4r + 6$

r	2	0	−3	9	−5	4
t	14					

d $z = 5y + 2$

y	4	8	7	−5	6	−2
z		42				

e $y = 2x - 3$

x	0	1	2	3	4	5
y						

f $b = \frac{1}{2}a - 2$

a	−2	−1	0	1	2	3
b						

g $k = 3h + 5$

h	−1	0	1	2	3	4
k						

h $y = 1 - x$

x	−2	−1	0	1	2	3
y						

4 Which table of values matches the formula $y = 3x - 1$? Select **A**, **B**, **C** or **D**.

A

x	1	2	3	4	5
y	2	3	4	5	6

B

x	1	2	3	4	5
y	2	5	8	11	14

C

x	1	2	3	4	5
y	4	7	10	13	16

D

x	1	2	3	4	5
y	0	3	6	9	12

5 Which table of values follows the rule $c = \frac{h+2}{2}$? Select **A**, **B**, **C** or **D**.

A

h	0	2	4
c	1	3	5

B

h	0	2	4
c	1	2	5

C

h	0	2	4
c	0	1	3

D

h	0	2	4
c	1	2	3

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Graphing linear equations 9.06

A **linear equation** is an equation or formula whose graph on a number plane is a **straight line**.

Graphing linear equations

- complete a table of values
- graph the table of values on a number plane
- rule a line through the points and label the line with the equation.

Example 2

Graph $y = x + 1$ on a number plane.

Solution

Complete a table of values. Choose x-values close to 0 for easy calculation and graphing.

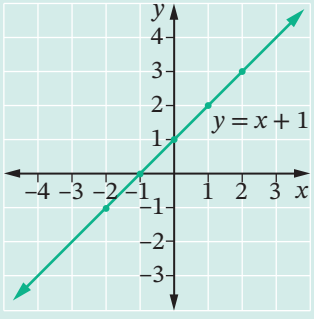
x	−2	−1	0	1	2
y	−1	0	1	2	3

Graph the table of values on a number plane.

Rule a straight line through the points, place arrows at each end, and label the line with its equation.

Note:

- Every pair of values that follows the linear equation $y = x + 1$ lies on this line, for example, (3, 4), $(-4, -3)$, $(1\frac{1}{2}, 2\frac{1}{2})$, (5, 6), (100, 101), $(-\frac{1}{2}, \frac{1}{2})$.
- Every point on the line follows the linear equation $y = x + 1$.
- There are an infinite number of points that satisfy the rule.



For these reasons, the line is infinite so we must draw arrows at each end.

Worksheets
Graphing linear equations
Number plane grid paper
A page of lines

Puzzles
Matching linear equations
Graphing functions

Interactives
Function builder: Basics
Function builder
Graphing lines

Technology
Graphing straight lines: Finding intercepts

Spreadsheet
Linear function plotter

Videos
Graphing linear equations
Linear functions

Example 3

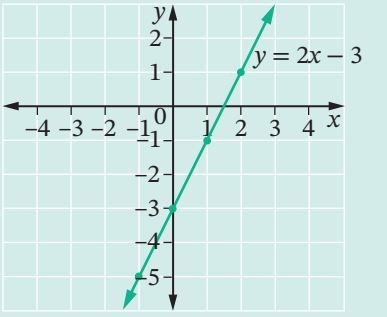
a Graph $y = 2x - 3$ on a number plane.

b Test whether the point $(5, 8)$ lies on the line $y = 2x - 3$.

Solution

a

x	−1	0	1	2
y	−5	−3	−1	1



b We could extend the line $y = 2x - 3$ to see if $(5, 8)$ lies on it, but a simpler way is to check whether $(5, 8)$ follows or **satisfies** the formula $y = 2x - 3$. If it does, then when $x = 5$, $y = 8$.

When $x = 5$, $y = 2 \times 5 - 3 = 7 \neq 8$

$\therefore$ $(5, 8)$ does not lie on the line. (In fact, $(5, 7)$ does)

EXERCISE 9.06 ANSWERS ON P. 593

Graphing linear equations UFRC

EXAMPLE 2

Worksheets
Number plane grid paper
A page of number planes

EXAMPLE 3

1 For each linear equation, copy and complete the table of values and graph the equation on a number plane. C

a $y = x + 3$

x	−1	0	1	2
y				

b $y = x - 2$

x	0	1	2	3
y				

c $y = 2x$

x	−1	0	1	2
y				

d $y = \frac{x}{2}$

x	−1	0	1	2
y				

e $y = 6 - x$

x	0	1	2	3
y				

f $y = x$

x	−1	0	1	2
y				

2 Graph each linear equation on a number plane.

a $y = 2x + 1$ **b** $y = 2x - 1$ **c** $y = 3x$

d $y = 3x - 2$ **e** $y = 2x + 4$ **f** $y = 3x - 5$

3 Look at the graphs you drew in question **2**. Why are they called **increasing graphs**? C

4 Graph each linear equation on a number plane.

a $y = -x$ **b** $y = -x + 5$ **c** $y = 4 - x$

d $y = -x - 3$ **e** $y = 1 - 3x$ **f** $y = -2x$

5 Look at the graphs you drew in question **4**. Why are they called **decreasing graphs**? C

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6 By looking at each linear equation, predict whether its graph will be increasing or decreasing. R C

a $y = 5x - 4$ **b** $y = x + 9$ **c** $y = 3 - x$

d $y = 2x$ **e** $y = -x + 5$ **f** $y = -3x - 1$

7 Test whether each point lies on the graph of the given linear equation by: R

i substituting its coordinates into the linear equation

ii examining the graphs of the lines you drew in questions **1** and **2**

a $(4, 7)\ y = x + 3$ **b** $(-1, -1)\ y = x - 2$

c $(5, 1)\ y = 6 - x$ **d** $(-3, -5)\ y = 2x + 1$

e $(4, 10)\ y = 3x - 2$ **f** $(-2, 1)\ y = 2x + 4$

8 On which line does the point $(1, 3)$ lie? Select the correct answer **A**, **B**, **C** or **D**.

A $y = -x + 3$ **B** $y = x + 4$ **C** $y = 2 - 2x$ **D** $y = 2x + 1$

9.06

Investigation

Height vs head circumference

Is your height related to the circumference of your head?

A number plane can be used to graph different types of information.



Alamy Stock Photo/Margot Hartford

1 Draw a table to collect data about the students from your class:

Name	Head circumference (C)	Height (H)

2 For each student in the class, measure (in cm):

- Height (without shoes)
- Circumference of head

3 Draw a number plane using the first quadrant only. Place Head circumference (C) on the horizontal axis and Height (H) on the vertical axis.

4 For each person, plot a point using your data.

5 By considering your graph, do you think there is a relationship between head circumference and height?

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Power plus ANSWERS ON P. 595

Worksheet
Plane grid 2

1 **a** Draw a number plane extending from –7 to 7 on the x-axis, and from 12 to –8 on the y-axis, or use the worksheet 'Plane grid 2'.

b Start at the first point and join the points in the order given.

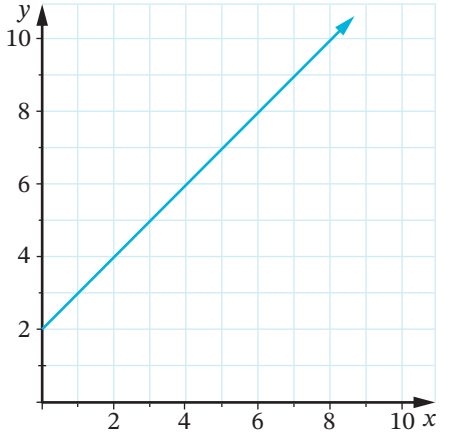
- $(3\frac{1}{2}, 0), (4, 2), (4, 4), (3\frac{1}{2}, 7), (4\frac{1}{2}, 8), (4\frac{1}{2}, 9), (5, 10), (4\frac{1}{2}, 11\frac{1}{2}), (3\frac{1}{2}, 10), (2, 10), (\frac{1}{2}, 11\frac{1}{2}), (1, 10), (1, 8), (1\frac{1}{2}, 7), (2, 6\frac{1}{2}), (1\frac{1}{2}, 4\frac{1}{2}), (0, 2\frac{1}{2}), (-2\frac{1}{2}, 1), (-3, 0), (-3\frac{1}{2}, -4), (-4\frac{1}{2}, 6), (-4, -7), (0, -8), (2, -7\frac{1}{2}), (2, -6\frac{1}{2}), (0, -7)$, STOP
- $(-2, -7\frac{1}{2}), (1, -6), (3, -6), (4, -6\frac{1}{2}), (4\frac{1}{2}, -5\frac{1}{2}), (3, -4\frac{1}{2}), (0, -5), (-3, -6\frac{1}{2})$, STOP
- $(\frac{1}{2}, -2), (1, -3), (0, -5)$, STOP
- $(1\frac{1}{2}, 0), (1, -2), (1\frac{1}{2}, -4\frac{1}{2})$, STOP
- $(2, -6), (2\frac{1}{2}, -7\frac{1}{2}), (3\frac{1}{2}, -7\frac{1}{2}), (3\frac{1}{2}, -6\frac{1}{2}), (3, -6)$, STOP
- $(2\frac{1}{2}, -4\frac{1}{2}), (2\frac{1}{2}, -2), (3\frac{1}{2}, 0)$, STOP
- $(1\frac{1}{2}, -3), (1, -5)$, STOP

c Complete the picture as you wish.

d What have you drawn?

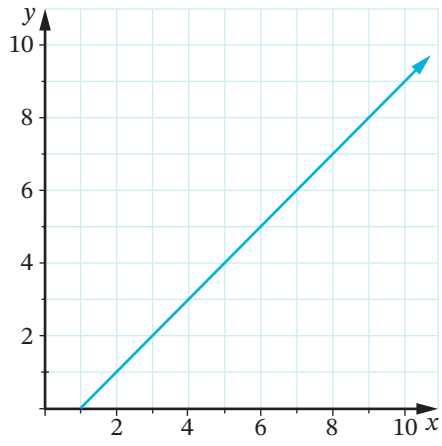
2 Complete each table by reading values from each graph. Find an algebraic formula $y =$ _____ that relates each y-value with the x-value.

a



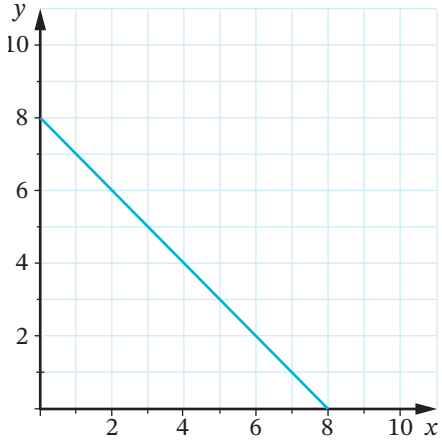
x	y
0	
1	
3	
6	
8	

b



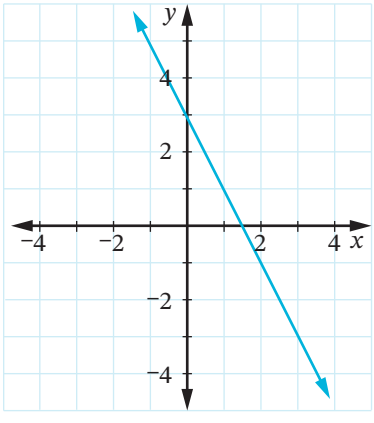
x	y
1	
4	
	6
8	
10	

c



x	y
0	
	6
	4
6	
	0

d



x	y
−1	
0	
	−1
3	
	−5

CHAPTER 9 REVIEW

Language of maths

Quiz
Language of maths 9

Puzzle
Number plane crossword

axis/axes	Cartesian	collinear	coordinates
linear equation	number plane	ordered pair	origin
point	quadrant	satisfy	table of values
x-axis	*y*-axis		

1 Look up the word 'coordinate' in the dictionary. Find both its mathematical and non-mathematical meanings.

2 Why do you think 'ordered pair' is another name for coordinates?

3 What is the **origin** and why does it have that name?

4 What word is used to describe a set of points 'lying in a line'?

5 What is another name for the number plane?

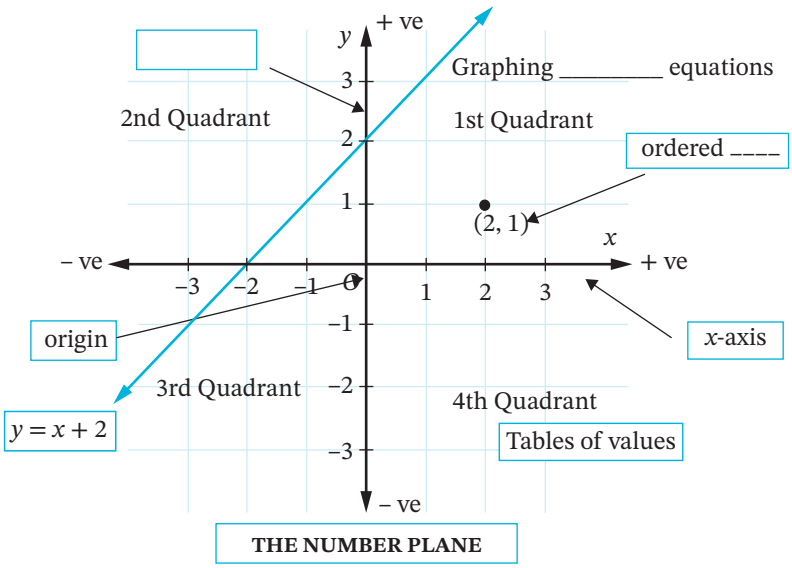
6 Why is $y = 4x + 1$ called a **linear equation**?

Topic summary

Worksheet
Mind map: The number plane

- In your own words, write about how points are plotted on the number plane.
- In your own words, write about how coordinates are used to locate a position on the number plane.
- What parts of this topic did you find difficult? What parts didn't you understand? Discuss them with a friend or your teacher.
- Give some examples of where the number plane is used.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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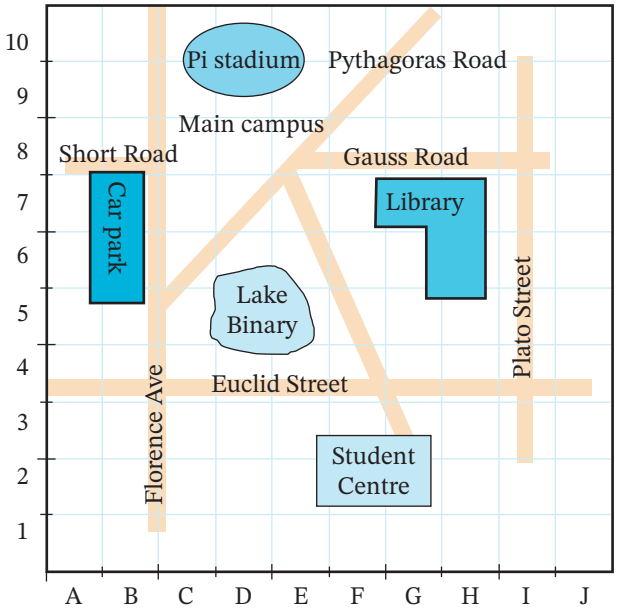
TEST YOURSELF 9

ANSWERS ON P. 596

Quiz
Test yourself 9

9.01

1 This is a map of the Cartesian University campus.



State the feature that appears at:

a F2 **b** D9 **c** A8 **d** H7

2 Use the map in question **1** to write the location of:

9.01

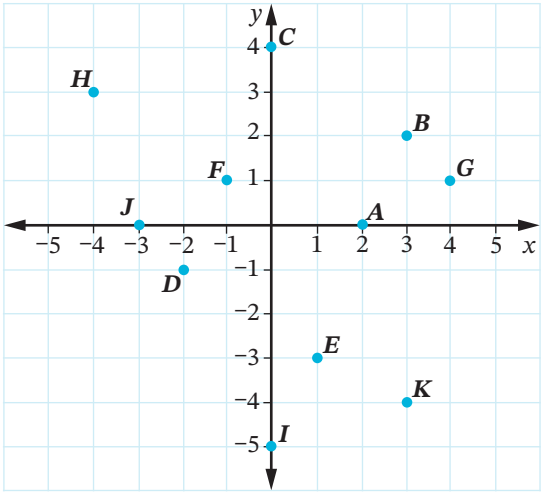
a Lake Binary **b** the car park **c** Gauss Rd

3 Plot each point on the same number plane.

9.02

$A(2, 3)$, $B(1, 6)$, $C(3, 0)$, $D(0, 4)$, $E(5, 2)$, $F(6, 5)$

4 Write the coordinates of each labelled point on this number plane.



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9.03

5 **a** In which quadrant is the point (−3, −3)?

b Write the coordinates of a point on the *y*-axis.

c In which quadrant is a point with a positive *x*-coordinate and a negative *y*-coordinate?

d Write the coordinates of a point in the 2nd quadrant.

e What are the coordinates of the origin?

9.03

6 In which quadrant is a point with a negative *x*-coordinate and a positive *y*-coordinate? Select the correct answer **A**, **B**, **C** or **D**.

A 1st **B** 2nd **C** 3rd **D** 4th

9.03

7 **a** Plot these points on a number plane, joining them with straight lines.

(1, 3), (2, 3), (2, 2), (3, 1), (2, 0), (2, −1), (1, −1), (0, −2), (−1, −1), (−2, −1), (−2, 0), (−3, 1), (−2, 2), (−2, 3), (−1, 3), (0, 4), (1, 3).

b What shape have you drawn?

9.04

8 Plot the following sets of points on a number plane. Use a ruler to join each set of points.

a

x	−2	−2	−2	−2
y	−2	−1	0	1

b

x	−2	−1	0	2
y	3	3	3	3

9.04

9 Graph each table of values on a separate number plane. Use a ruler to join up the points and describe their pattern.

a

x	*y*
0	5
1	4
2	3
5	0

b

x	−1	0	1	2
y	−1	1	3	5

9.05

10 Copy and complete each table of values.

a $y = x + 3$

x	−1	0	1	2	3
y					

b $q = 2p - 7$

p	−2	−1	0	1	2
q					

c $d = 5 - c$

c	−1	0	1	2	3
d					

9.06

11 Graph each linear equation on a number plane after completing a table of values.

a $y = 2x - 1$ **b** $y = -x + 3$ **c** $y = \frac{1}{2}x - 2$

9.06

12 Test whether each point lies on the line $y = 3x + 2$.

a (1, 2) **b** (−1, 4) **c** (−2, −4) **d** (0, 2)

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PRACTICE SET 3

ANSWERS ON P. 597

1 Mark and label each point on a number plane. 9.03

A (1, 3) B (–2, 2) C (3, –4)

D (–2, –3) E (0, 7) F (–3, 0)

2 Draw a number line labelled from 0 to 3, and mark each decimal on it. 7.01

0.5 2.25 1.3 2.9 0.75 1.6

3 How many metres in 735 centimetres? Select the correct answer **A, B, C** or **D.** 8.01

A 73 500 **B** 7.35 **C** 73.5 **D** 7350

4 Evaluate each expression. 7.03

a 1.54 + 7.9 **b** 9.2 – 6.1 **c** 22.6 – 13.55

d 6.31 + 0.2 + 15.38 **e** 0.003 + 1.01 + 4.394 **f** 9.803 – 4.52

5 Convert each fraction to a decimal. 7.09

a $\frac{1}{5}$ **b** $\frac{3}{8}$

6 Copy and complete each conversion. 8.01

a 2000 mg = _____ g **b** 2 t = _____ kg **c** 56 days = _____ weeks

d 6.5 L = _____ mL **e** 3 days = _____ hours **f** 4 kg = _____ g

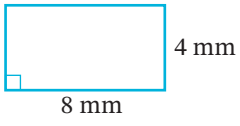
g 1500 g = _____ kg **h** 3000 mL = _____ L **i** 480 s = _____ min

j 2500 kg = _____ t **k** 9.5 g = _____ mg **l** 12 000 000 mg = _____ kg

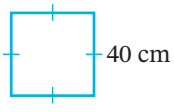
7 A sheet of A4 paper has a width of 21.1 cm and a length of 29.7 cm. Find its perimeter. 8.02

8 Find the perimeter of each shape. 8.02

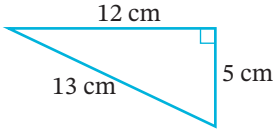
a



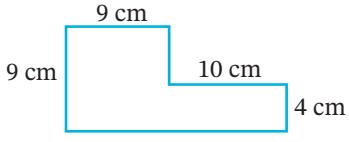
b



c



d



9 Find the area of each shape above. 8.08

10 Round: 7.10

a 0.753 to one decimal place **b** 12.064 to 2 decimal places

c $9.8217 to the nearest cent **d** 3.795 to the nearest tenth

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7.01

11 Which letter represents 3.3 kg on this number line? Select **A**, **B**, **C** or **D**.



8.13

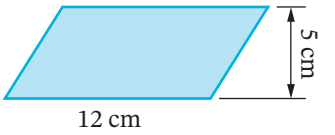
12 Convert 7500 L to:

a kL **b** m^3

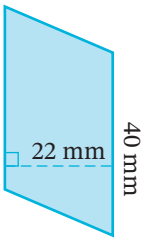
8.07

13 Find the area of each parallelogram.

a



b



9.04

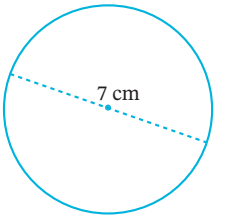
14 Graph this table of values on a number plane, with both axes extending from –4 to 4. Use a ruler to join the points.

x	–1	0	2	3
y	–4	–2	2	4

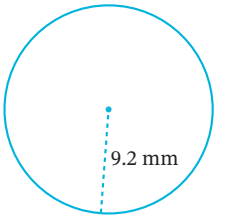
8.03

15 Find the circumference of each circle, correct to one decimal place.

a



b



9.06

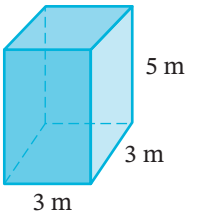
16 For the linear equation $y = 3 - x$, copy and complete this table of values and graph the equation on a number plane.

x	–1	0	1	2
y				

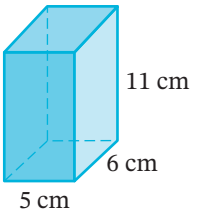
8.11

17 Find the volume of each rectangular prism.

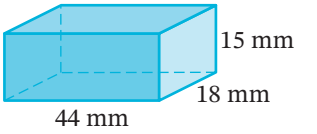
a



b



c



7.09

18 Convert $\frac{5}{9}$ to a recurring decimal.

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19 **a** If the temperature in Brisbane was 20.6°C at 9 a.m. and was 29.3°C at 3 p.m., by how much did it rise? 7.11

b A piece of cloth 2.5 m long was cut into 10 equal pieces. How long was each piece?

c Libby bought 40 litres of LPG at 79.9c per litre. Linton bought 40 litres at 87.9c per litre. How much more than Libby did Linton pay?

20 Which number is the largest? Select **A**, **B**, **C** or **D**. 7.01

A 0.83 **B** $\frac{4}{5}$ **C** $\frac{81}{100}$ **D** 0.82

21 Find the capacity of a box measuring 48 cm by 33 cm by 17 cm 8.13

a in mL **b** in L

22 Evaluate each product. 7.06

a 6.2×0.7 **b** 4.21×0.3 **c** 0.05×1.4

23 Evaluate each quotient. 7.07

a $60.138 \div 6$ **b** $1.672 \div 0.02$ **c** $4.032 \div 0.3$

24 **a** A rectangle has an area of 20 cm^2. If its length is 5 cm, how wide is it? 8.05

b A square has an area of 64 cm^2. Find its perimeter.

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